



AIS trajectories & port calls

2024 fleet · 7,256 containerships · 674,055 legs

A · DATA CONSTRUCTION



Vessel & fuel parameters

engine power · speed · fuel type ·
WtW factors



Voyage-leg emission estimation

AIS cleaning · port-call reconstruction · WtW accounting
216.2 Mt CO₂e at leg level



H3 spatial attribution

Res 5 open ocean · Res 7 chokepoints · emission-conserving
cell routes
1,181,711 hexagonal cells · 216.2 Mt CO₂e



Corridor construction & feasibility

OD aggregation · fuel range · intermediate bunkering chains
7,769 corridors · 206.4 Mt CO₂e (95.4% coverage)



Governance inputs

EU ETS · FuelEU · BRI scopes ·
budgets · carbon prices



Governance attribution classifier

EU ETS endpoint rule · BRI membership · FuelEU pressure
field
priority: EU ETS > BRI > FuelEU > vacuum · port camps

C · RULE COMPETITION



Rule-competition MILP family

M^G global planner · M^{EU} EU rule sphere · M^{BRI} invest-
ment domain
shared decision space · budgets · ETS revenue recycling

D · POLICY EVIDENCE



Policy experiments & dynamic extension

ϵ -constraint frontier · hub control · market activation · 2025–
2050 sequencing
indices: Rule-Vacuum · Mismatch Cost · Hub Control

RQ1

Where do rules reach?

Rule-sphere coverage, overlap, and
high-emission rule-vacuum lanes

RQ2

What does fragmentation cost?

Competition utility matrix and pol-
icy mismatch cost across rule spheres

RQ3

Who controls the transition?

Hub control across camps, break-even
carbon prices, and investment timing